SAMPLE ON PAPER

SESSION 2025-2026

PHYSICS

Class 11th

SET-01

Time: 120 Minutes

M.M.: 80

General Instructions:

1. Answers to this Paper must be written on the paper provided separately.
2. You will not be allowed to write during first 15 minutes.
3. This time is to be spent in reading the question paper.
4. The time given at the head of this Paper is the time allowed for writing theanswers.
5. Section A is compulsory. Attempt any four questions from Section B.
6. The intended marks for questions or parts of questions are given in brackets [ ].

SECTION-A

(Attempt all questions from this Section.)

1. In questions (i) to (vii) below, choose the correct alternative (a), (b), (c), or (d) for each of the questions given below:

(i) Convert 5 N to Dynes.

(A) 500000 Dynes
(B) 500 Dynes
(C) 5 Dynes
(D) 0.5 Dynes

(ii) What kind of motion is rectilinear motion?

(A) One-dimensional
(B) Two-dimensional
(C) Three-dimensional
(D) Zero-dimensional

(iii) Which of the following is not true about projectile motion?

(A) It is an example of motion in a plane
(B) It is an example of motion along a curve
(C) It is not an example of motion in space
(D) The acceleration keeps changing in projectile motion

(iv) How much should a spring of indefinite length be compressed to have a potential energy equivalent to a ball of mass 6 kg raised to a height of 120 m above the ground? Let the spring have a stiffness of k = 100 N/m and assume g = 10 m/s^2.

(A) 6m (B) 12m
(C) 20m (D) 144m

(v) In 0.02800 number of significant figures will be :

(A) 2 (B) 3
(C) 4 (D) 5

(vi) If the kinetic energy of a satellite is halved, its orbital radius becomes _____

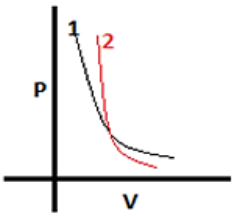
(A) 1/2 times

(B) 2 times

(C) 4 times

(D) infinity

(vii) In the given diagram, one graph is of an ideal gas and another is of a real gas. Select the correct option.



(A) 1-real gas, 2-ideal gas

(B) 1-ideal gas, 2-real gas

(C) Both are for an ideal gas at different temperatures

(D) Their graphs cannot intersect

(B) Answer the following questions briefly:

(i) What is meant by weightlessness? Explain with an example.

(ii) Give the limitations of Dimensional analysis.

(iii) The magnitudes of two vectors A and B are 5 and 4 units, and the angle between them is 30°. Find the dot product of two vectors.

(iv) How many times larger is kg than mg?

(v) What is the angle between the direction of velocity and acceleration at the highest point of the projectile's path?

(vi) A force of 40 N acts on a body for 10 sec. What will be the change in momentum

(vii) Name two physical quantities having dimensions the same as work.

SECTION B – 14 MARKS

2. Find the dimensions of a and b in the relation

$P = (b - x^2)/at$, where P is pressure, x is distance, and t is time.

OR

Write down Kepler's laws of planetary motion. Prove that the force acting on a planet is inversely proportional to the square of its distance from the sun.

3. A body covers a distance of (13.8 ± 0.2)m in time (4.0 ± 0.3) scc. Calculate its speed with error limits.

4. Explain two applications of impulse.

OR

The length, breadth, and thickness of the metal sheet are 4.234 m. 1.005 m and 2.01 cm respectively. Give the area and the volume of the sheet to correct significant figures.

5. Explain what will be the value of the vertical component of velocity and the highest point of projectile motion.

6. If the lift is going up with a speed of 2 m/s and the mass of a man standing on it is 60kg, find its apparent weight.

7. A car travels the first half of the distance between two places with a speed of 40 km/hr and the rest half distance with a speed of 50 km/hr. What is the average speed of a car?

8. Prove that the horizontal range is the same when the angle of projection is (i) greater than 45° by a certain value and (ii) less than 45° by the same value.

[2]

SECTION C – 27 MARKS

9. A 10 g moving body is acted upon by a force of 10 N for 3 milliseconds. Find the impulse and the change in velocity of the body.

OR

A string can bear a max tension of 72 kgf. By what acceleration can a man descend by this rope? The mass of the man is 96000 g.

10. Find the angle between two vectors:
$=2i + 2j - k$ & $B = 6i - 3j + 2k$

11. What should be the two possible angles of projections if a body is thrown with 100 m/sec and covers a horizontal distance of half a km?

12. What will be the value of the force of 100 N in a system having meter, Kg, and minute as fundamental units?

13. Draw a velocity-time graph and represent uniform velocity and uniform acceleration on it. What does the area under the velocity-time graph represent?

14. If a ball is dropped from the top of a tower, it takes 10 seconds to reach the ground. Find the distance travelled in 5th sec.

15. Draw a graph between :

(i) Linear momentum and velocity when mass is constant.

(ii) Velocity and mass if momentum is constant.

OR

A 20 g ball hits a wall at 45° with a speed of 54 km/hr and rebounds back perpendicular to the direction of incidence with the same speed. Find the impulse received by the ball.

16. Find the angle between two equal vectors such that the magnitude of the resultant is equal to either of the two vectors.

17. A closed pipe and an open-ended pipe produce simultaneously a sound with 5 beats per second. If the length of the pipe is 30cm, then determine the required change in the length of the closed pipe, if both pipes are to produce the sound of same frequency. The speed of sound is 330m/s.

SECTION D – 15 MARKS

18. The frequency f of an oscillatory liquid drop depends on the radius r of the drop, density d, and surface tension s. Obtain the formula of frequency f by dimension.

OR

A source is sound revolving in a circle of radius 15m is emitting a signal of frequency 200Hz. It completes one revolution in 3 seconds. Calculate the maximum and minimum frequencies of the signal heard at a point 30 m from the centre of the circle. The speed of sound is 330m/s.

19. The temperature inside the refrigerator is –3ºC and the room temperature is 27ºC. How many joules of heat will be delivered to the room for each joule of electrical energy consumed ideally?

OR

A ball is thrown from the top of the tower with an initial velocity of 10 m/s at an angle of 30° above the horizontal. It hits the ground at a distance of 17.3 m from the base of the tower. Calculate the height of the tower. ($g = 10\ m/s^2$)

20. Derive three equations of motion by the Calculus & Graphical method.

Useful Constants and Relations:

1. $g = 9.8\ m/s^2$.
2. Density of water $= 10^3\ kg/\ m^3$.

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